

GrounUp Enterprise Specification

Phase 29 - Enterprise Reporting, Data Warehouse/Lakehouse, Semantic Analytics & AI-Ready Data

Version 1.0.0 | Implementation-ready specification

1. Mission

Implement a governed enterprise analytics platform that converts operational GrounUp data into secure, historical, certified and reusable data products for reporting, forecasting, benchmarking and AI-ready features.

2. Security Inheritance

Phase 29 inherits Phase 28 identity, tenant isolation, row/column security, classification, privacy, encryption, audit and secure service-identity controls. Analytics cannot become a back door around operational permissions.

3. Warehouse and Lakehouse

Raw/bronze, validated/silver and curated/gold layers preserve source metadata and transformation versions. Conformed warehouse facts and dimensions provide stable historical analytical grain.

4. ETL/ELT and Orchestration

CDC, batch and streaming pipelines use versioned contracts, checkpoints, retries, idempotency, backfills and run evidence.

5. Quality, Reconciliation and Lineage

Certified assets require data-quality, freshness and reconciliation controls plus source-to-metric/report lineage. Blocking failures prevent certification or trigger controlled stale/incident state.

6. Semantic Model and Metrics

Certified measures/KPIs define expression, grain, filters, units, time basis, owner and version. Official dashboards and reports use certified definitions.

7. Reporting and Self-Service

Role-based dashboards, scheduled reports, exports and self-service exploration operate through permission-aware semantic models.

8. Forecasting and Benchmarking

Forecasts, scenarios and benchmarks are versioned, labeled and separated from actual results. Cross-tenant benchmarks obey privacy/aggregation policy.

9. AI-Ready Analytics

Feature snapshots and embedding metadata preserve source/version/security lineage and point-in-time correctness for AI evaluation, forecasting and future learning systems.

10. Implementation Order

Source contracts -> ingestion -> raw/validated/curated layers -> quality/reconciliation/lineage -> warehouse -> semantic metrics -> marts -> reports/dashboards -> forecasting -> AI-ready features.

Analytics Modules

ID	Module	Layer	Purpose
P29-MOD-001	Analytics Control Plane	Analytics Platform	Dataset, pipeline, model, metric, report and ownership registry
P29-MOD-002	Operational Replication Service	Data Integration	Read-only governed replication/CDC from operational domains
P29-MOD-003	Batch Ingestion Service	Data Integration	Scheduled batch ingestion with checkpoints/retries/evidence
P29-MOD-004	Streaming Ingestion Service	Data Integration	Near-real-time event ingestion with ordering/idempotency controls
P29-MOD-005	Raw/Bronze Data Zone	Lakehouse	Immutable/raw analytical landing with source metadata
P29-MOD-006	Validated/Silver Data Zone	Lakehouse	Cleaned/conformed data with quality/reconciliation state
P29-MOD-007	Curated/Gold Data Zone	Lakehouse	Business-ready curated datasets and dimensional outputs
P29-MOD-008	Warehouse Fact Service	Warehouse	Fact table definitions, loads, partitioning and historical grain
P29-MOD-009	Dimension Service	Warehouse	Conformed dimensions and slowly changing dimension handling
P29-MOD-010	Transformation Registry	ETL/ELT	Versioned SQL/code/model transformations and dependencies
P29-MOD-011	Pipeline Orchestrator	ETL/ELT	Schedules, dependencies, retries, backfills, SLAs and run state
P29-MOD-012	Schema/Data Contract Registry	Governance	Source/target schemas, compatibility and owner contracts
P29-MOD-013	Data Quality Engine	Governance	Completeness, validity, uniqueness, consistency and anomaly checks
P29-MOD-014	Reconciliation Engine	Governance	Source-to-analytics counts/amounts/control-total reconciliation
P29-MOD-015	Freshness Monitor	Operations	Data recency/latency SLAs and stale-data alerting
P29-MOD-016	Lineage Service	Governance	Column/dataset/report/model source-to-output lineage graph
P29-MOD-017	Metadata Catalog	Governance	Searchable business/technical metadata, owners and certifications
P29-MOD-018	Semantic Model Service	Analytics	Governed dimensions, relationships, measures and query semantics
P29-MOD-019	Metric/KPI Registry	Analytics	Certified metric formulas, grain, filters, owner and version
P29-MOD-020	Time Intelligence Service	Analytics	Calendar/fiscal/period comparisons and historical windows
P29-MOD-021	Estimating Analytics Mart	Domain Analytics	Estimate, line, production, price, margin and win/loss analytics
P29-MOD-022	Project/Job Cost Mart	Domain Analytics	Budget/actual/commitment/forecast/production analytics
P29-MOD-023	Schedule Analytics Mart	Domain Analytics	Milestone, float, performance, resource and forecast analytics
P29-MOD-024	Commercial Analytics Mart	Domain Analytics	Contract/change/claim/payment/exposure analytics
P29-MOD-025	CRM/Sales Analytics Mart	Domain Analytics	Lead/opportunity/customer conversion and pipeline analytics
P29-MOD-026	Workforce Analytics Mart	Domain Analytics	Labor, crew, skills, utilization and capacity analytics
P29-MOD-027	Fleet Analytics Mart	Domain Analytics	Equipment utilization, cost, maintenance, fuel and downtime analytics
P29-MOD-028	Procurement Analytics Mart	Domain Analytics	Vendor, PO, price, lead-time and commitment analytics
P29-MOD-029	Safety Analytics Mart	Domain Analytics	Incident, inspection, action and trend analytics
P29-MOD-030	AI/Document Analytics Mart	Domain Analytics	AI usage/evals, document intelligence, review and evidence analytics
P29-MOD-031	Regional Benchmark Mart	Analytics	Regional rates, production, cost and benchmark distributions
P29-MOD-032	Reporting Service	Reporting	Governed reports, parameters, formats and report execution
P29-MOD-033	Dashboard Service	Reporting	Role-based operational/executive dashboards
P29-MOD-034	Self-Service Analytics Service	Reporting	Permission-aware ad hoc exploration using certified semantic model
P29-MOD-035	Subscription/Export Service	Reporting	Scheduled reports, subscriptions and controlled exports
P29-MOD-036	Forecasting Service	Advanced Analytics	Time-series/driver-based forecasts with versioned assumptions

ID	Module	Layer	Purpose
P29-MOD-037	Scenario Analytics Service	Advanced Analytics	Scenario comparison across cost/production/schedule/commercial drivers
P29-MOD-038	AI-Ready Feature Store	AI Data	Governed analytical features with source/version/freshness metadata
P29-MOD-039	Embedding Metadata Registry	AI Data	Vector/embedding provenance and source/version permissions metadata
P29-MOD-040	Analytics Security Service	Security	Tenant/row/column security and privacy enforcement
P29-MOD-041	Analytics Observability Service	Operations	Pipeline/query/report quality, performance and cost telemetry
P29-MOD-042	Analytics Incident Service	Operations	Data incident triage, impact, backfill, correction and evidence

Candidate Certified Metrics

ID	Metric	Formula	Domain	Unit
P29-KPI-001	Estimate Hit Rate	won_bids / submitted_bids	Estimate/CRM	%
P29-KPI-002	Estimate Gross Margin %	(sell_value - estimated_cost) / sell_value	Estimating	%
P29-KPI-003	Job Cost Variance %	(actual_cost - budget_cost) / budget_cost	Project/Finance	%
P29-KPI-004	Production Attainment %	actual_production / planned_production	Operations	%
P29-KPI-005	Schedule Performance Index	earned_value / planned_value	Scheduling	ratio
P29-KPI-006	Equipment Utilization %	productive_hours / available_hours	Fleet	%
P29-KPI-007	Crew Utilization %	assigned_productive_hours / available_hours	Workforce	%
P29-KPI-008	Change Exposure	approved_change + pending_change + forecast_change	Commercial	currency
P29-KPI-009	Procurement On-Time %	on_time_deliveries / due_deliveries	Procurement	%
P29-KPI-010	Safety Action Closure %	closed_on_time_actions / due_actions	HSE	%
P29-KPI-011	AI Grounding Pass %	grounded_pass_cases / evaluated_cases	AI Governance	%
P29-KPI-012	Data Freshness Compliance %	datasets_within_sla / certified_datasets	Analytics	%